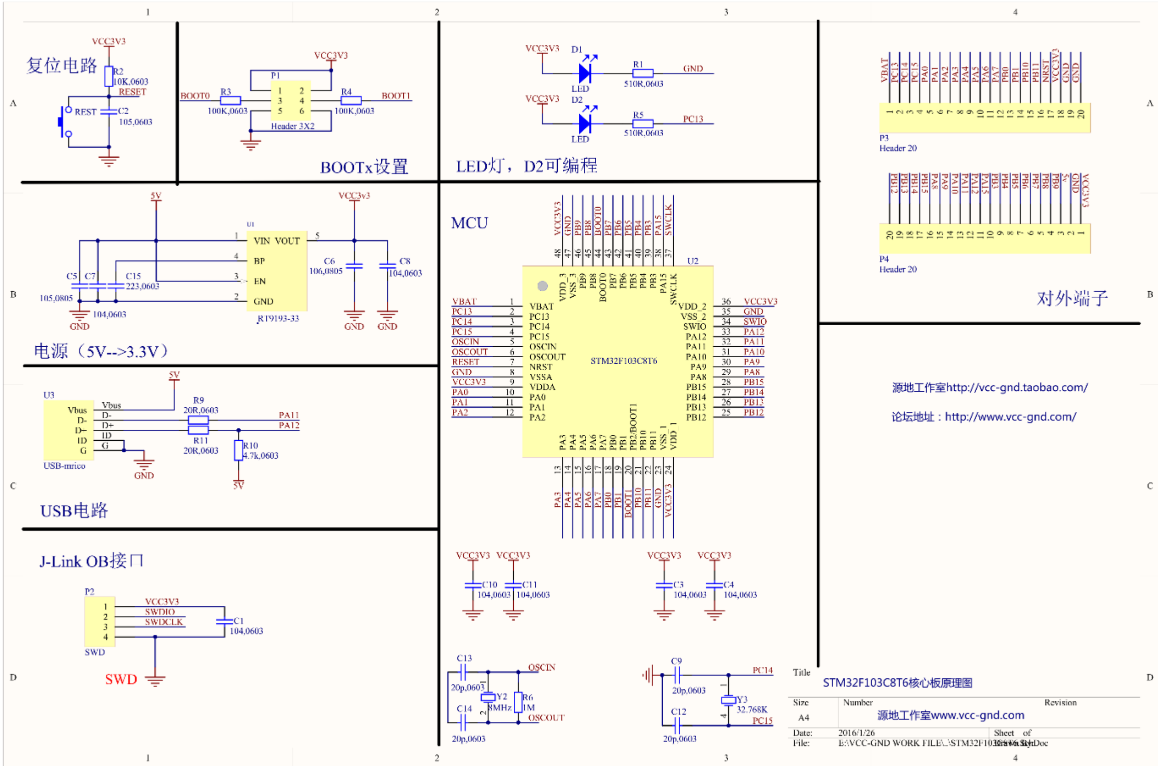
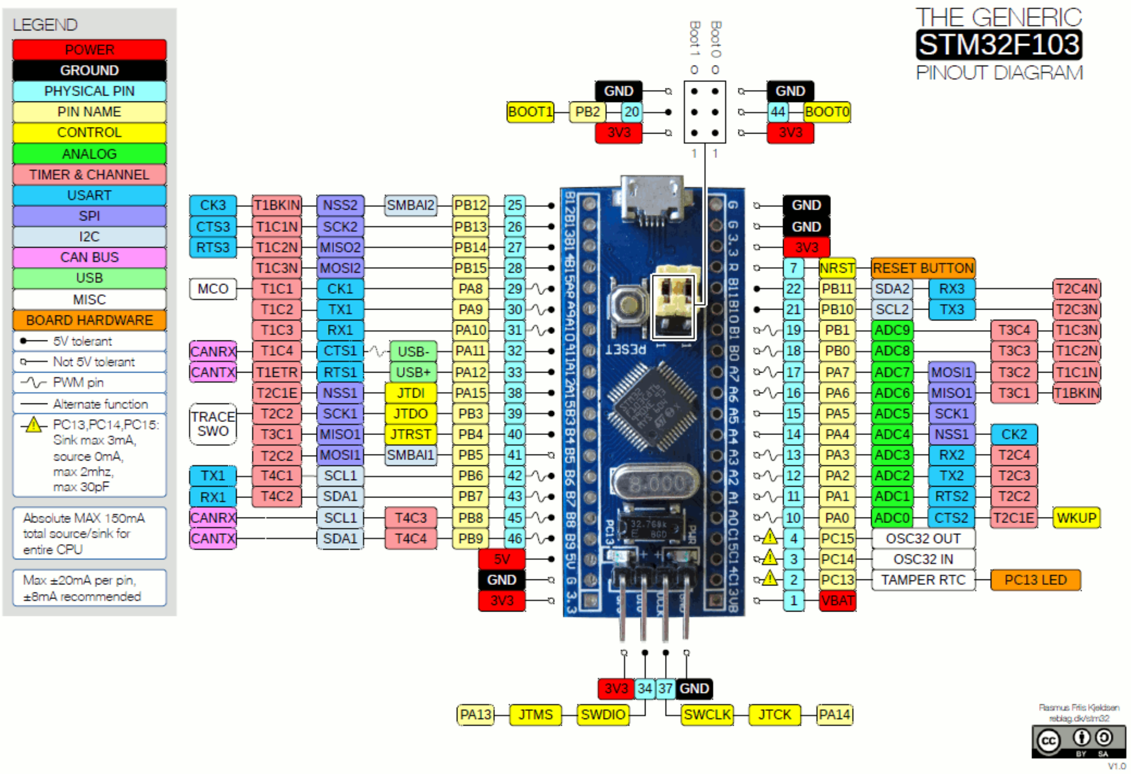
Chapter 2
Digital and Analog I/O

Link: https://www.electronicshub.org/getting-started-with-stm32f103c8t6-blue-pill/
Link: https://electropeak.com/learn/how-to-program-stm32-blue-pill-stm32f103c8t6-with-arduino-ide/
Link: https://stm32-base.org/assets/pdf/boards/original-schematic-STM32F103C8T6-Blue_Pill.pdf
Datasheet:
https://www.st.com/resource/en/datasheet/stm32f103c8.pdf
Reference Manual:
https://www.st.com/resource/en/reference_manual/rm0008-stm32f101xx-stm32f102xx-stm32f103xx-stm32f105xx-and-stm32f107xx-advanced-armbased-32bit-mcus-stmicroelectronics.pdf


复位电路
BOOTx设置
LED灯，D2可编程
MCU
STM32F103C8T6
对外端子
电源（5V-->3.3V）
USB电路
J-Link OB接口
SWD
源地工作室http://vcc-gnd.taobao.com/
论坛地址：http://www.vcc-gnd.com/
STM32F103C8T6核心板原理图
源地工作室www.vcc-gnd.com


THE GENERIC
STM32F103
PINOUT DIAGRAM
LEGEND
POWER
GROUND
PHYSICAL PIN
PIN NAME
CONTROL
ANALOG
TIMER & CHANNEL
USART
SPI
I2C
CAN BUS
USB
MISC
BOARD HARDWARE
5V tolerant
Not 5V tolerant
PWM pin
Alternate function
PC13,PC14,PC15: Sink max 3mA, source 0mA, max 2mhz, max 30pF
Absolute MAX 150mA total source/sink for entire CPU
Max ±20mA per pin, ±8mA recommended
RESET BUTTON
OSC32 OUT
OSC32 IN
TAMPER RTC
PC13 LED
Rasmus Friis Kjeldsen
reblag.dk/stm32
V1.0

Experiment 1 - Blinking LED

```
void setup() {
  // put your setup code here, to run once:
  pinMode(PC13, OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(PC13, HIGH);
  delay(1000);
  digitalWrite(PC13, LOW);
  delay(1000);
}
```

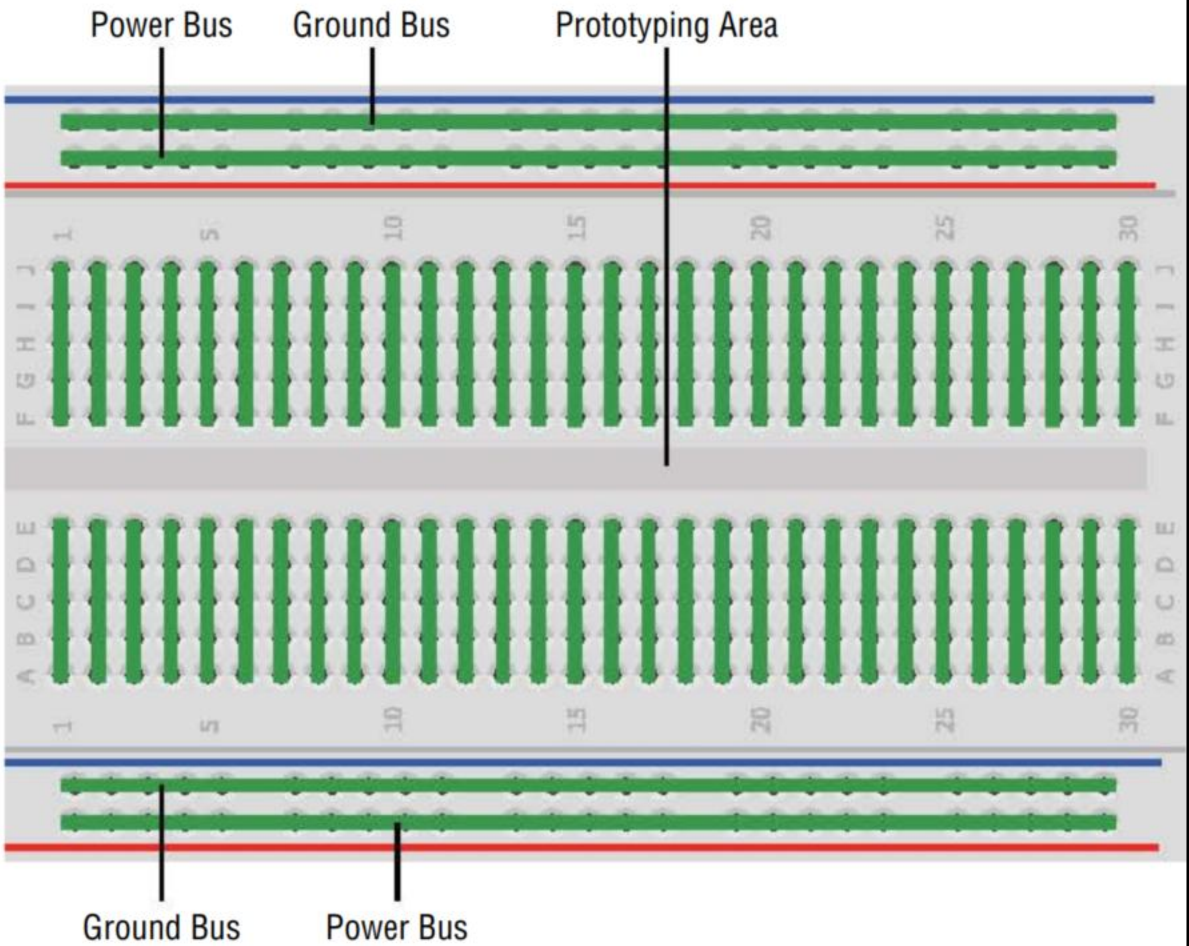


Figure 2-1: Breadboard electrical connections

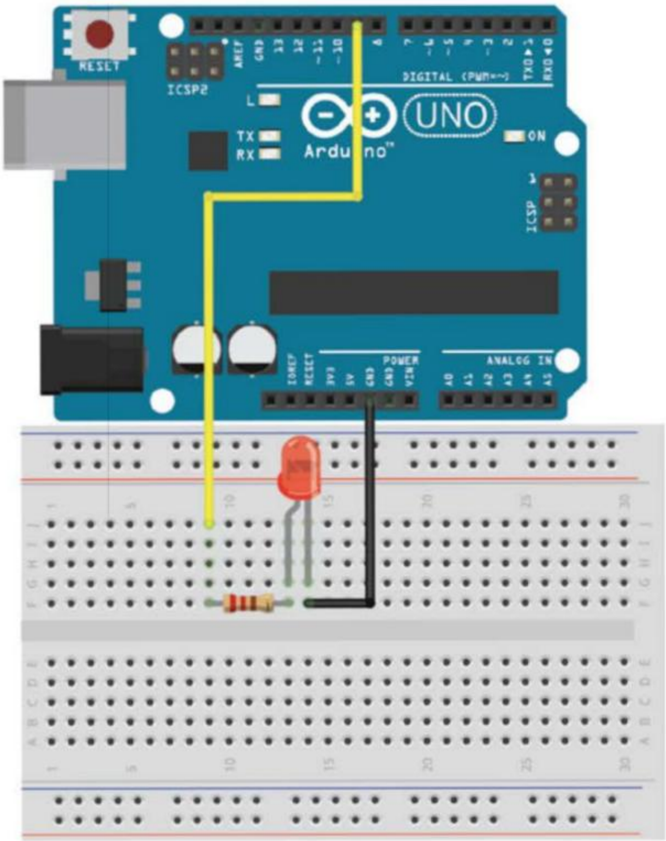


Figure 2-2: Arduino Uno wired to an LED

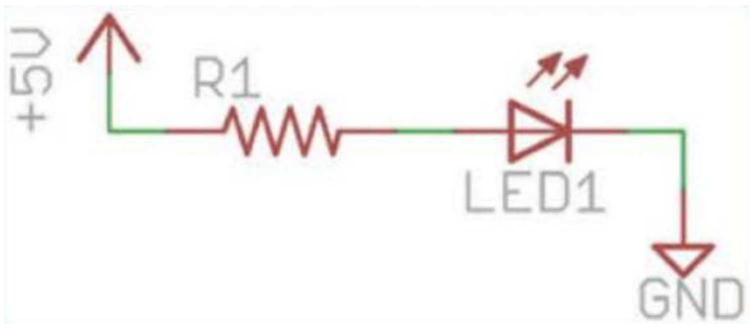


Figure 2-3: Simple LED circuit

220Ω resistor

List of some of the digital output pins commonly used on the Blue Pill:

1. PA0 - PA15 (Port A, Pins 0-15)
2. PB0 - PB15 (Port B, Pins 0-15)
3. PC13 - PC15 (Port C, Pins 13-15)

List of digital output pins available on the Arduino Uno:

1. **Digital Pins 0 to 13:** These pins can be used as digital input or output pins. Digital Pins 0 and 1 are also connected to the hardware serial interface (RX and TX), respectively.
2. **Digital Pins A0 to A5:** These pins can be used as digital input or output pins, in addition to their analog input functionality.

Experiment 2 - Turning On An LED

```
void setup()
{
   pinMode (PA15, OUTPUT);
   // Set the LED pin as an output
   digitalWrite(PA15, HIGH);
   // Set the LED pin high
}

void loop()
{
   // We are not doing anything in the loop!
}
```

Experiment 3 - LED with Changing Blink Rate

```
void setup()
{
   pinMode (PA15, OUTPUT);
   // Set the LED pin as an output
}

void loop()
{
   for (int i=100; i<=1000; i=i+100)
   {
       digitalWrite(PA15, HIGH);
       delay(i);
       digitalWrite(PA15, LOW);
       delay(i);
   }
}
```

Pulse-Width Modulation with analogWrite()

List of analog output (PWM) pins commonly used on the Blue Pill:

1. PA0 (Timer2 Channel 1)
2. PA1 (Timer2 Channel 2)
3. PA2 (Timer2 Channel 3)
4. PA3 (Timer2 Channel 4)
5. PA6 (Timer3 Channel 1)
6. PA7 (Timer3 Channel 2)
7. PB0 (Timer3 Channel 3)
8. PB1 (Timer3 Channel 4)
9. PB6 (Timer4 Channel 1)
10. PB7 (Timer4 Channel 2)

List of digital output pins available on the Arduino Uno:

1. Digital Pins 3, 5, 6, 9, 10, and 11: These pins support PWM functionality and can be used for analog output via PWM.

Experiment 4 - Fading LED

```
void setup()
{
   pinMode (PB0, OUTPUT);
   //Set the LED pin as an output
}

void loop()
{
  for (int i=0; i<256; i++)
  {
     analogWrite(PB0, i);
     delay(10);
  }
  for (int i=255; i>=0; i--)
  {
     analogWrite(PB0, i);
     delay(10);
  }
}
```

Reading Digital Inputs

List of digital input pins commonly used on the Blue Pill:

1. PA0 - PA15 (Port A, Pins 0-15)
2. PB0 - PB15 (Port B, Pins 0-15)
3. PC0 - PC15 (Port C, Pins 0-15)

List of digital input pins available on the Arduino Uno:

1. Digital Pins 0 to 13: These pins can be used as digital input or output pins. Digital Pins 0 and 1 are also connected to the hardware serial interface (RX and TX), respectively.
2. Digital Pins A0 to A5: These pins can be used as digital input or output pins, in addition to their analog input functionality.

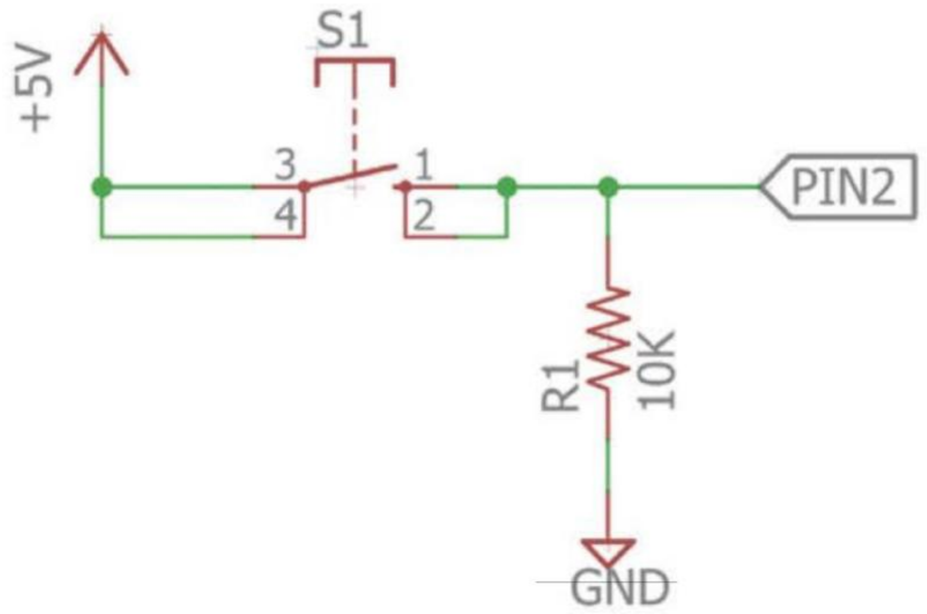


Figure 2-5: Pushbutton input with pull-down resistor schematic

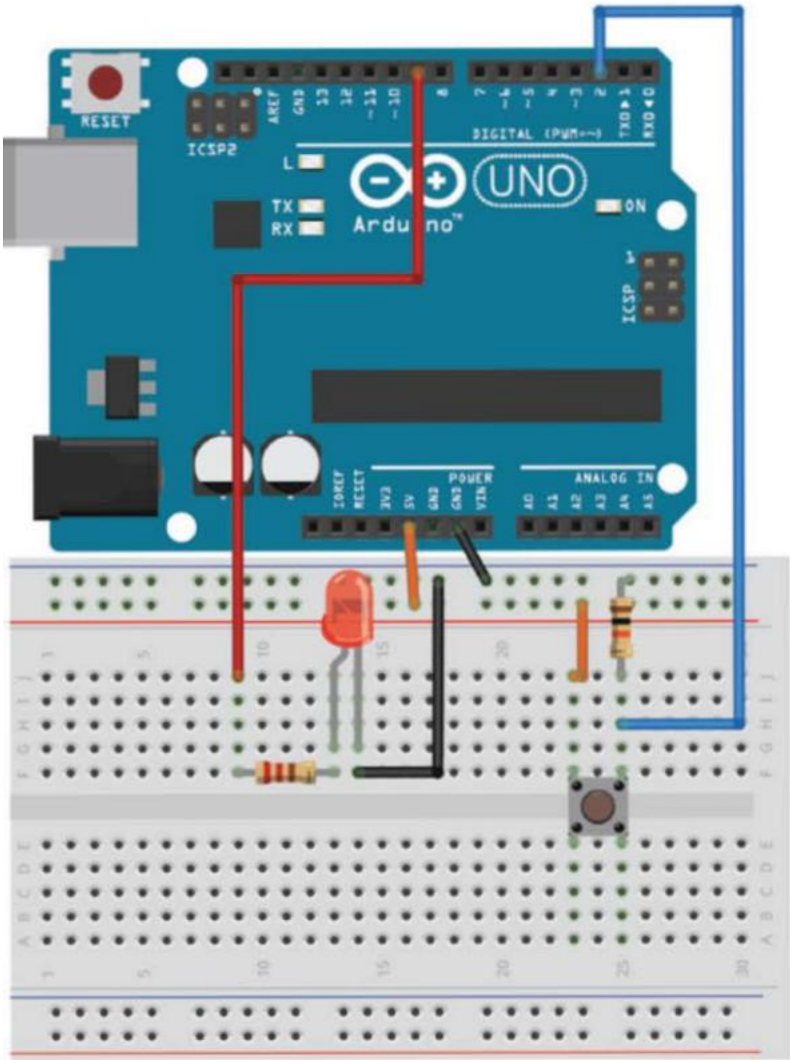


Figure 2-6: Wiring an Arduino to a button and an LED

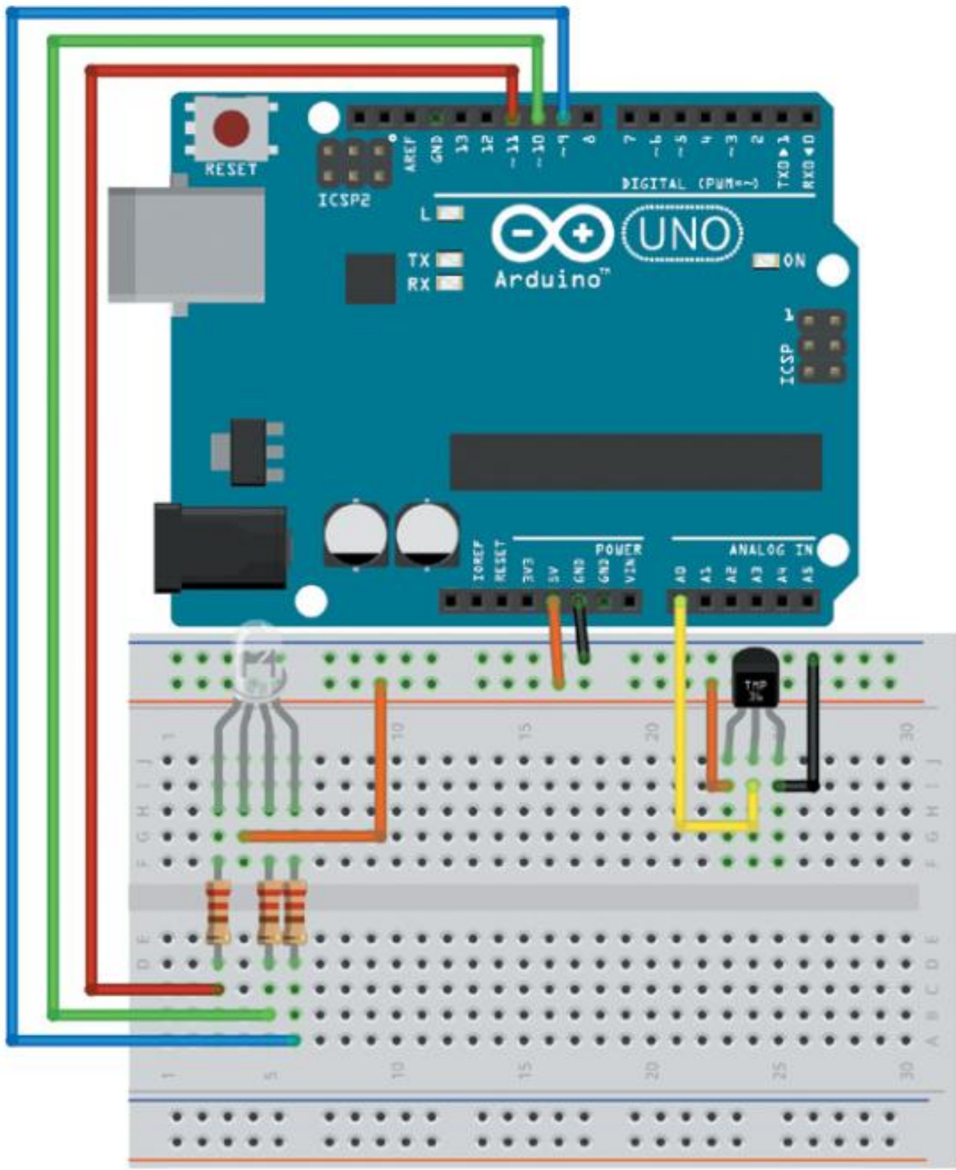
Experiment 5 - Simple LED Control With a Button

```
const int LED=PB0;
// The LED is connected to pin 9
const int BUTTON=PC15;
// The Button is connected to pin 2

void setup()
{
   pinMode (LED, OUTPUT);
   // Set the LED pin as an output
   pinMode (BUTTON, INPUT);
   // Set button as input (not required)
}

void loop()
{
   if (digitalRead(BUTTON) == LOW)
   {
      digitalWrite(LED, LOW);
    }
    else
    {
      digitalWrite(LED, HIGH);
    }
}
```

Experiment 6 - Reading temperature sensor



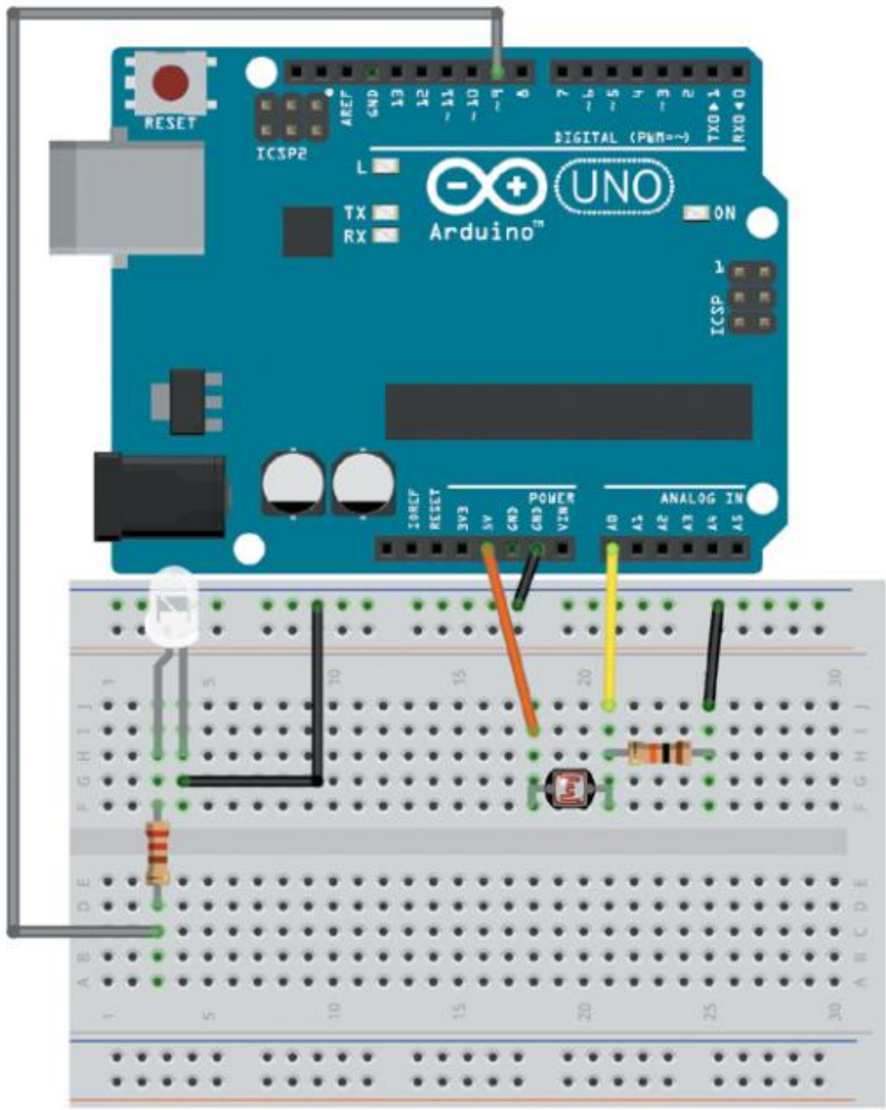

```
// Temperature Alert!
const int BLED=PB0; // Blue LED Cathode on
Pin B0
const int GLED=PB1; // Green LED Cathode on
Pin B1
const int RLED=PB2; // Red LED Cathode on Pin
B2
const int TEMP=PA0; // Temp Sensor is on pin
```

```
A0
const int LOWER_BOUND=139; // Lower Threshold
const int UPPER_BOUND=147; // Upper Threshold
int val = 0; // Variable to hold analog
reading

void setup() {
    pinMode (TEMP, INPUT);
    pinMode (BLED, OUTPUT); // Set Blue LED
    as Output
    pinMode (GLED, OUTPUT); // Set Green LED
    as Output
    pinMode (RLED, OUTPUT); // Set Red LED as
    Output
}
void loop() {
    val = analogRead(TEMP);
    if (val < LOWER_BOUND) { // LED is Blue
        digitalWrite(RLED, HIGH);
        digitalWrite(GLED, HIGH);
        digitalWrite(BLED, LOW);
    }
    else if (val > UPPER_BOUND) { // LED is
    Red
        digitalWrite(RLED, LOW);
        digitalWrite(GLED, HIGH);
        digitalWrite(BLED, HIGH);
    }
```

```
    else { // LED is Green
        digitalWrite(RLED, HIGH);
        digitalWrite(GLED, LOW);
        digitalWrite(BLED, HIGH);
    }
}
```

Experiment 7 - Reading photoresistor




```
// Automatic Night Light
const int WLED=PA0; // White LED Anode on pin
```

```
PA0 (PWM)
const int LIGHT=PA1; // Light Sensor on
Analog Pin PA1
const int MIN_LIGHT=200; // Minimum Expected
light value
const int MAX_LIGHT=900; // Maximum Expected
Light value
int val = 0; // Variable to hold the analog
reading

void setup() {
    pinMode(WLED, OUTPUT); // Set White LED
pin as output
}
void loop() {
    val = analogRead(LIGHT); // Read the
    light sensor
    val = map(val, MIN_LIGHT, MAX_LIGHT, 255,
    0); // Map the light reading
    val = constrain(val, 0, 255); //
    Constrain light value
    analogWrite(WLED, val); // Control the
    White LED
}
```

Course Plan & Teachers

Nahin Ul Sadad, *Assistant Professor, CSE, RUET*

Md. Farhan Shakib, *Lecturer, CSE, RUET*

Md. Nasif Osman Khansur, *Lecturer, CSE, RUET*